

RAG Week Assignment – Section A: Vector Databases

A1. What is a Vector Database?

Question

Explain in your own words what a vector database is and how it's different from a traditional database like MySQL or PostgreSQL.

- A vector database is a specialized database used to store embeddings (vectors).
- These embeddings represent the meaning of text, images, audio, or other data.
- Traditional databases store structured data and are optimized for exact matches.
- Vector databases are optimized for similarity search.
- Instead of finding the exact same value, they find the most similar information.
- This makes them useful for AI applications such as RAG, recommendation systems, and semantic search.

A2. Raw Text to Vector Database

Question

Describe what happens from raw text to storage in a vector database.

Example Sentence: "Artificial Intelligence helps machines learn from data."

- Step 1: Raw text is received.
- Step 2: The text is cleaned and prepared.
- Step 3: Tokenization breaks the sentence into smaller units.
- Step 4: An embedding model converts the sentence into a numerical vector.
- Step 5: Metadata such as source or document ID is added.
- Step 6: The vector and metadata are stored in the vector database.
- Step 7: The database creates indexes for fast similarity search.

A3. Why Not Use a Normal Database Index?

Question

Explain the difference between exact lookup and similarity search.

- B-tree indexes work well for exact matches and sorted values.
- Embeddings contain hundreds or thousands of dimensions.
- Similarity search compares the meaning of vectors instead of exact values.
- Two different sentences can have similar embeddings even if the words differ.
- Traditional indexes are not designed for high-dimensional similarity calculations.
- Vector databases use specialized algorithms such as HNSW and IVF for efficient retrieval.

A4. Cosine Similarity Calculation

A = [1, 2, 3]

B = [2, 4, 6]

- Dot Product = $(1 \times 2) + (2 \times 4) + (3 \times 6) = 28$
- $\|A\| = \sqrt{1^2 + 2^2 + 3^2} = \sqrt{14}$
- $\|B\| = \sqrt{2^2 + 4^2 + 6^2} = \sqrt{56}$
- Cosine Similarity = $28 / (\sqrt{14} \times \sqrt{56})$
- Since $\sqrt{56} = 2\sqrt{14}$, denominator = 28
- Cosine Similarity = $28 / 28 = 1$

Result:

- The cosine similarity is 1.0.
- This means both vectors point in exactly the same direction.
- Therefore, they are perfectly similar.

A5. Why is a Vector Database Important in RAG?

- RAG systems need to retrieve relevant information before generating answers.

- User queries are converted into embeddings.
- The vector database finds the most relevant chunks using similarity search.
- Retrieved information is then sent to the LLM.
- Without a vector database, retrieval would be slow and less accurate.
- The model may miss important context and generate incorrect answers.
- Therefore, the vector database acts as the knowledge retrieval layer of a RAG system.